Roll's Theorem & Taylor's Theorem & Maclaurin's Series

2022 [Total marks = 37]

Q. 1 (b) The function $f(x) = x(x+3)e^{-\frac{x}{2}}$ satisfy all condition of Roll's mean value theorem in the interval $[-3, 0]$. Then the value of c is :

(a) 0 (b) 1 (c) 2 (d) -2

$\Rightarrow$

Given; $f(x) = x(x+3)e^{-\frac{x}{2}}$, interval $[-3, 0]$

$$f(x) = [x^2 + 3x]e^{-\frac{x}{2}}$$

To verify Roll's theorem;

① $f(a) = f(b)$

$$f(-3) = [(-3)^2 + 3\times(-3)]e^{-\frac{3}{2}}$$

$$f(-3) = [9 - 9]e^{-\frac{3}{2}}$$

$$f(-3) = 0 \quad \text{— (i)}$$

$$f(0) = [0^2 + 3\times 0]e^{-\frac{0}{2}}$$

$$f(0) = 0 \quad \text{— (ii)}$$

from eq-(i) and (ii) we can say

$$f(-3) = f(0)$$

$\therefore$ its satisfy the condition; $f(a) = f(b)$.

$f'(c) = 0$ such that $c \in (a,b)$.

$f(c) = [c^2 + 3c]\, e^{-\frac{c}{2}}$

$f'(c) = e^{-c/2}\,[2c+3] - \frac{1}{2} e^{-\frac{c}{2}}\,[c^2+3c]$

[applying $\frac{d}{dx}[uv]$]

$f'(c) = e^{-c/2}\left[2c+3 - \frac{[c^2+3c]}{2}\right]$

$f'(c) = e^{-c/2}\left[\frac{4c+6-c^2-3c}{2}\right]$

$f'(c) = \frac{1}{2e^{c/2}}\left[c+6-c^2\right]$

$f'(c) = 0$

$\frac{1}{2e^{c/2}}\left[c+6-c^2\right] = 0$

$c+6-c^2 = 0$

$c^2 - c - 6 = 0$

$c^2 - 3c + 2c - 6 = 0$

$c(c-3) + 2(c-3) = 0$

$(c-3)(c+2) = 0$

$c = 3$ or -2

at $c = 3$

$c \in (a,b)$

$3 \notin (-3,0)$

at $c = -2$

$-2 \in (-3,0)$

$\therefore$ $\boxed{\text{the value of } c \text{ is } -2}$

Q.1.(i) If $f(x)$ is continous in the closed interval $[a,b]$ and $f(x)$ exists in (a,b) and $f(a) = f(b)$, then there exist at least one value c $(a < c < b)$ such that $f'(c) = 0$ is called

(i) Taylor's theorem
(ii) Maclaurin's theorem
(iii) Rolle's theorem
(iv) Lagrange's mean value theorem

$\Rightarrow$

Rolle's theorem;

1. function should be continous and differentiable
2. $f(a) = f(b)$
3. $f'(c) = 0$ such that $c \in (a,b)$ or $(a < c < b)$

Q.4.(a) Obtain the fourth-degree Taylor's polynomial approximation to $f(x) = e^{2x}$ about $x = 0$. Find the maximum error when $0 \le x \le 0.5$.

$\Rightarrow$ Given, $n = 4$

$f(x) = e^{2x}$ about $x = 0$

error when $0 \le x \le 0.5$.

Taylor's series;

$$f(x+a) = f(a) + x f'(a) + \frac{x^2}{2!} f''(a) + \frac{x^3}{3!} f'''(a) + \frac{x^4}{4!} f^{iv}(a)$$

here ; Taylor expansion of e^{2x} at $a=0$

Rough
$f(x+a) = f(x)$
$f(x+0) = f(x)$
$x = x$
$a = 0$

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \frac{x^4}{4!} f^{iv}(0).$$

$f(x) = e^{2x}$

$f(0) = e^{2\times 0} = e^0 = 1$ $\quad [e^0 = 1]$

$f'(x) = 2e^{2x}$

$f'(0) = 2e^0 = 2$

$f''(x) = 4e^{2x}$

$f''(0) = 4e^0 = 4$

$f'''(x) = 8e^{2x}$

$f'''(0) = 8e^0 = 8$

$f^{iv}(x) = 16e^{2x}$

$f^{iv}(0) = 16e^0$

$f^{iv}(0) = 16$

$f^{v}(x) = 32e^{2x}$

$f^{v}(0.5) = 32e^{2\times 0.5}$

$= 32e^1$

$f^{v}(0.5) = 32e$

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \frac{x^4}{4!} f^{iv}(0)$$

$$f(x) = 1 + x(2) + \frac{x^2}{2}(4) + \frac{x^3}{6}(8) + \frac{x^4}{24}(16)$$

$$\boxed{f(x) = 1 + 2x + 2x^2 + \frac{4}{3}x^3 + \frac{2}{3}x^4}$$

Hence, this is a fourth-degree taylor expansion of $f(x) = e^{2x}$ about $x = 0$.

Now, to find maximum error ;

~~Remaind~~

Error form $R_n = \frac{x^{n+1}}{(n+1)!} f^{n+1}(x+\theta a)$

at $n=4$

$$R_4 = \frac{x^{4+1}}{(4+1)!} f^{4+1}(x+\theta a)$$

$$R_4 = \frac{x^5}{5!} f^5(x+\theta a)$$

Here $a=0$

$0 \le x \le 0.5$

at $x=0$

Remainder $(R_4)=0$

at $x=0.5$

$$R_4 = \frac{(0.5)^5}{5!} f^5(0.5+0)$$

$$R_4 = \frac{\left(\frac{1}{2}\right)^5}{120} f^5(0.5)$$

$$R_4 = \frac{1}{120 \times 2^5} \times 32\, e = \frac{e}{120}$$

Hence, $\boxed{\text{maximum error} = \frac{e}{120}}$

(Q1) If $f(x) = \log(1+x)$, $x>0$ using maclaurin's theorem, show that for $0<\theta<1$.

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3(1+\theta x)^3}$$

Deduce that

$$\log(1+x) < x - \frac{x^2}{2} + \frac{x^3}{3} \quad \text{for } x>0.$$

Given; $f(x) = \log(1+x)$

Maclaurin's series;

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) \cdots\cdots \frac{x^{n-1}}{(n-1)!} f^{n+1}(\theta x)$$

$f(x) = \log(1+x)$

$f(0) = \log(1+0)$

$= \log(1) = 0$

$f'(x) = \frac{1}{1+x}$

$f'(0) = \frac{1}{1+0} = 1$

$f''(x) = \frac{-1}{(1+x)^2}$

$f''(0) = \frac{-1}{(1+0)^2} = -1$

$$f'''(x) = \frac{2}{(1+x)^3}$$

$$f'''(\theta x) = \frac{2}{(1+\theta x)^3}$$

(for $n = 2$ with remainder form)

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(\theta x)$$

$$= 0 + x(1) + \frac{x^2}{2}(-1) + \frac{x^3}{6}\left(\frac{2}{(1+\theta x)^3}\right)$$

$$\boxed{f(x) = x - \frac{x^2}{2} + \frac{x^3}{3(1+\theta x)^3}}$$ proved.

Hence this is the maclaurin series of

$$f(x) = \log(1+x).$$

Now;

As per question given;

$0 < \theta < 1$ and $x > 0$

Here, $\theta > 0$ and $x > 0$
So we can write
$\theta x > 0$

$\theta x > 0$

add 1 both side

$1+\theta x > 1$

$(1+\theta x)^3 > 1^3$

$1 > \dfrac{1}{(1+\theta x)^3}$

multiply $\dfrac{x^3}{3}$ both side

$\dfrac{x^3}{3} > \dfrac{x^3}{3(1+\theta x)^3}$

add $\left(x - \dfrac{x^2}{2}\right)$ both side

$x - \dfrac{x^2}{2} + \dfrac{x^3}{3} > x - \dfrac{x^2}{2} + \dfrac{x^3}{3(1+\theta x)^3}$

$x - \dfrac{x^2}{2} + \dfrac{x^3}{3} > \log(1+x)$ $\left[\log(1+x) = x - \dfrac{x^2}{2} + \dfrac{x^3}{3(1+\theta x)^3}\right]$

$$\boxed{\therefore \log(1+x) < x - \frac{x^2}{2} + \frac{x^3}{3}}$$

proved.

Q.6. (b) Using Taylor's theorem, express the polynomial $2x^3+7x^2+x-6$ in power of $(x-1)$.

$\Rightarrow$ Given;

$$f(x) = 2x^3+7x^2+x-6 \text{ in power of } (x-1)$$

Taylor's series;

$$f(x+a) = f(a) + x f'(a) + \frac{x^2}{2!} f''(a) + \frac{x^3}{3!} f'''(a) + \frac{x^4}{4!} f^{iv}(a) + \cdots$$

Rough

$f(x+a) = f(x)$

$f((x-1)+1) = f(x)$

$x = x-1$

$a = 1$

$$f(x) = f(1) + (x-1) f'(1) + \frac{(x-1)^2}{2!} f''(1) + \frac{(x-1)^3}{3!} f'''(1) + \frac{(x-1)^4}{4!} f^{iv}(1) + \cdots$$

$\cancel{f(x) = 2x^3+7x^2+x-6}$

$\cancel{f(0) = 2\times(0)^3+7(0)^2+0-6}$

$\cancel{f(0) = -6}$

$\cancel{f'(x) = 6x^2+14x+1}$

$\cancel{f'(0) =}$

$f(x) = 2x^3+7x^2+x-6$

$f(1) = 2(1)^3+7(1)^2+1-6$

$f(1) = 4$

$f'(x) = 6x^2+14x+1$

$f'(1) = 6(1)^2+14(1)+1$

$f'(1) = 21$

$f''(x) = 12x + 14$

$f''(1) = 12(1) + 14$

$f''(1) = 26$

$f'''(x) = 12$

$f'''(1) = 12$

$f^{iv}(x) = 0$

$f^{iv}(1) = 0$

(Similarly till $f^{\infty}(x) = 0$)

$$f(x) = f(1) + (x-1)f'(1) + \frac{(x-1)^2}{2!}f''(1) + \frac{(x-1)^3}{3!}f'''(1) + \frac{(x-1)^4}{4!}f^{iv}(1) + \cdots$$

$$f(x) = 4 + (x-1)(21) + \frac{(x-1)^2}{2}(26) + \frac{(x-1)^3}{6}(12) + \frac{(x-1)^4}{24}(0) + \cdots$$

$$\boxed{f(x) = 4 + 21(x-1) + 13(x-1)^2 + 2(x-1)^3 \cdots}$$

Hence, this is the taylor series expansion of $f(x) = 2x^3 + 7x^2 + x - 6$ in power of $(x-1)$.

Q.8.(a).) Expand $\log x$ in power of $(x-1)$ by taylor's theorem.

$\Rightarrow$ Given;

$f(x) = \log x$ in the power of $(x-1)$

Taylor's series;

$$f(x+a) = f(a) + x f'(a) + \frac{x^2}{2!} f''(a) + \frac{x^3}{3!} f'''(a) + \dots$$

Rough

$f(x+a) = f(x)$

$f((x-1)+1) = f(x)$

$x = x-1$

$a = 1$

$$f(x) = f(1) + (x-1) f'(1) + \frac{(x-1)^2}{2!} f''(1) + \frac{(x-1)^3}{3!} f'''(1) + \dots$$

$f(x) = \log(x)$

$f(1) = \log(1) = 0$

$f'(x) = \frac{1}{x}$

$f'(1) = \frac{1}{1} = 1$

$f''(x) = \frac{-1}{x^2}$

$f''(1) = \frac{-1}{(1)^2} = -1$

$f'''(x) = \frac{+2}{x^3}$

$f'''(1) = \frac{2}{(1)^3} = 2$

$$f(x) = f(1) + (x-1)f'(1) + \frac{(x-1)^2}{2!}f''(1) + \frac{(x-1)^3}{3!}f'''(1) + \cdots$$

$$= 0 + (x-1)(1) + \frac{(x-1)^2}{2}(-1) + \frac{(x-1)^3}{6}(2) + \cdots$$

$$\boxed{f(x) = (x-1) - \frac{1}{2}(x-1)^2 + \frac{1}{3}(x-1)^3 + \cdots}$$

Hence, this is the taylor's expansion of $f(x) = \log x$ in the power of $(x-1)$.

Rolle's theorem

2022

4)(b) It is given the Rolle's theorem holds the function $f(x) = x^3 + bx^2 + cx$, $1 \le x \le 2$ at the point $x = 4/3$. Find the value of b and c.

$\Rightarrow$ Given;

$f(x) = x^3 + bx^2 + cx$, interval $[1,2]$

point $x = \frac{4}{3} \in [1,2]$

To verify Roll's theorem;

① $f(a) = f(b)$

$f(1) = (1)^3 + b(1)^2 + c(1)$

$f(1) = 1 + b + c$ —— (i)

$f(2) = (2)^3 + b(2)^2 + c(2)$

$f(2) = 8 + 4b + 2c$ —— (ii)

from eq- (i) and (ii) we get

$f(1) = f(2)$

$1 + b + c = 8 + 4b + 2c$

$3b + c + 7 = 0$ —— (iii)

② ~~f(x)~~ $f'(z) = 0$ such that $z \in (a,b)$

$f(z) = z^3 + bz^2 + cz$

$f'(z) = 3z^2 + 2bz + c$

$f'(z) = 0$

$3z^2 + 2bz + c = 0$

[(at $z = 4/3$) given]

$3\left(\frac{4}{3}\right)^2 + 2b\left(\frac{4}{3}\right) + c = 0$

$\frac{16}{3} + \frac{8b}{3} + c = 0$

$16 + 8b + 3c = 0$ —— (iv)

~~Subtract eq-(iii) by (iv) we get,~~

~~$(16 + 8b + 3c) - (3b + c + 7) = 0$~~

~~$5b + 2c$~~

from eq-(iii) and (iv)

$3b + c + 7 = 0$

$8b + 3c + 16 = 0$

multiply eq-(iii) by 3 and subtract from eq-(iv) we get,

$(8b + 3c + 16) - 3(3b + c + 7) = 0$

$8b + 3c + 16 - 9b - 3c - 21 = 0$

$-b - 5 = 0$

$b = -5$

put the value of b in eq- (iii)

$3(-5)+c+7=0$

$c=8$

Hence the value of $\boxed{\begin{array}{l} b=-5 \\ c=8 \end{array}}$

2023-24 [Expected Question]

Q. 1. If $f(h) = f(0) + hf'(0) + \frac{h^2}{2!} f''(\theta h)$, $0 < \theta < 1$, find θ when $h = 1$ and $f(x) = (1-x)^{5/2}$.

$\Rightarrow$ Given;

$$f(x) = (1-x)^{5/2}$$
$$\theta = ?$$

$$f(h) = f(0) + h f'(0) + \frac{h^2}{2!} f''(\theta h)$$

at $h = 1$

$$f(1) = f(0) + 1 f'(0) + \frac{1^2}{2!} f''(\theta)$$

$$f(1) = f(0) + f'(0) + \frac{1}{2} f''(\theta)$$

$$f(x) = (1-x)^{5/2}$$
$$f(0) = (1-0)^{5/2} = 1$$
$$f(1) = (1-1)^{5/2} = 0$$
$$f'(x) = -\frac{5}{2}(1-x)^{3/2}$$

$$f'(0) = -\frac{5}{2}(1-0)^{3/2} = -\frac{5}{2}$$

$$f''(x) = \frac{15}{4}(1-x)^{1/2}$$

$$f''(\theta) = \frac{15}{4}(1-\theta)^{1/2}$$

$$f(1) = f(0) + f'(0) + \frac{1}{2} f''(\theta)$$

$$0 = 1 + \left(-\frac{5}{2}\right) + \frac{1}{2}\left[\frac{15}{4}(1-\theta)^{1/2}\right]$$

$$0 = 1 - \frac{5}{2} + \frac{15}{8}\left[(1-\theta)^{1/2}\right]$$

$$= -\frac{3}{2} + \frac{15}{8}\left[(1-\theta)^{1/2}\right]$$

$$\frac{3}{2} = \frac{15}{8}(1-\theta)^{1/2}$$

$$\frac{8 \times 3}{2 \times 15} = (1-\theta)^{1/2}$$

$$\frac{4}{5} = (1-\theta)^{1/2}$$

$$\left(\frac{4}{5}\right)^2 = \left[(1-\theta)^{1/2}\right]^2$$

$$1-\theta = \frac{16}{25}$$

$$\theta = 1 - \frac{16}{25}$$

$$\boxed{\theta = \frac{9}{25}}$$

Hence the value of $\theta = \frac{9}{25}$ $(0<\theta<1)$.

Q.2. Using Maclaurin's series expand $\log(1+x)$. Hence deduce that

$$\log\sqrt{\frac{1+x}{1-x}} = x + \frac{x^3}{3} + \frac{x^5}{5} + \cdots$$

$\Rightarrow$ Given;

$$f(x) = \log(1+x)$$

Maclaurin's series

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \cdots$$

$$f(x) = \log(1+x)$$
$$f(0) = \log(1+0)$$
$$f(0) = \log(1) = 0$$
$$f'(x) = \frac{1}{1+x}$$
$$f'(0) = 1$$
$$f''(x) = \frac{-1}{(1+x)^2}$$
$$f''(0) = -1$$
$$f'''(x) = \frac{2}{(1+x)^3}$$
$$f'''(0) = 2$$

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \cdots$$
$$= 0 + x(1) + \frac{x^2}{2}(-1) + \frac{x^3}{6}(2) + \cdots$$

$$f(x) = x - \frac{x^2}{2} + \frac{x^3}{3} + \dots$$

Hence this is the Maclaurin's series expansion of $\log(1+x)$.

Now;

$$\log\sqrt{\frac{1+x}{1-x}}$$

$$\log\left(\frac{1+x}{1-x}\right)^{1/2}$$

$$\frac{1}{2}\left[\log\left(\frac{1+x}{1-x}\right)\right]$$

$$\frac{1}{2}\left[\log(1+x) - \log(1-x)\right]$$

Now for $\log(1+x)$
Maclaurin's series;

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \frac{x^4}{4!} f^{iv}(0) + \frac{x^5}{5!} f^{v}(0)$$

$$+ \dots$$

$$f(x) = \log(1+x)$$

$$f(0) = \log(1+0) = 0$$

$$f'(x) = \frac{1}{1+x}$$

$$f'(0) = 1$$

$$f''(x) = \frac{-1}{(1+x)^2}$$

$$f''(0) = -1$$

$$f'''(x) = \frac{2}{(1+x)^3}$$

$$f'''(0) = \frac{2}{(1+0)^3} = 2$$

$$f^{iv}(x) = \frac{-6}{(1+x)^4}$$

$$f^{iv}(0) = \frac{-6}{(1+0)^4} = -6$$

$$f^{v}(x) = \frac{24}{(1+x)^5}$$

$$f^{v}(0) = \frac{24}{(1+0)^5} = 24$$

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \frac{x^4}{4!} f^{iv}(0) + \frac{x^5}{5!} f^{v}(0) + \cdots$$

$$f(x) = 0 + x(1) + \frac{x^2}{2}(-1) + \frac{x^3}{6}(2) + \frac{x^4}{24}(-6) + \frac{x^5}{120}(24) + \cdots$$

$$f(x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} + \cdots \quad \text{—(i)}$$

This is the maclaurin's series of $\log(1+x)$

for $\log(1-x)$

Maclaurin's series:

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \frac{x^4}{4!} f^{iv}(0) + \frac{x^5}{5!} f^{v}(0) + \cdots$$

$f(x) = \log(1-x)$

$f(0) = \log(1-0) = 0$

$f'(x) = \frac{-1}{1-x}$

$f'(0) = \frac{-1}{1-0} = -1$

$f''(x) = \frac{-1}{(1-x)^2}$

$f''(0) = -1$

$f'''(x) = \frac{-2}{(1-x)^3}$

$f'''(0) = -2$

$f^{iv}(x) = \frac{-6}{(1-x)^4}$

$f^{iv}(0) = -6$

$f^{v}(x) = \frac{-24}{(1-x)^5}$

$f^{v}(0) = -24$

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \frac{x^4}{4!} f^{iv}(0) + \frac{x^5}{5!} f^{v}(0) + \cdots$$

$$f(x) = 0 + x(-1) + \frac{x^2}{2}(-1) + \frac{x^3}{6}(-2) + \frac{x^4}{24}(-6)$$

$$+ \frac{x^5}{120}(-24) + \cdots$$

$$f(x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \frac{x^5}{5} - \cdots \quad \text{--- (ii)}$$

This is the maclaurin's series of $\log(1-x)$

Now subtract eq-(ii) from (i) we get,

$$\log(1+x) - \log(1-x) = \left[x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} + \cdots\right] -$$

$$\left[-x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \frac{x^5}{5} - \cdots\right]$$

$$= \left[x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} \cdots + x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \frac{x^5}{5} + \cdots\right]$$

$$\log(1+x) - \log(1-x) = 2x + \frac{2x^3}{3} + \frac{2x^5}{5} + \cdots$$

$$\log(1+x) - \log(1-x) = 2\left[x + \frac{x^3}{3} + \frac{x^5}{5} + \cdots\right]$$

$$\frac{1}{2}\log(1+x) - \log(1-x) = x + \frac{x^3}{3} + \frac{x^5}{5} + \cdots$$

$$\boxed{\log\sqrt{\frac{1+x}{1-x}} = x + \frac{x^3}{3} + \frac{x^5}{5} + \cdots}$$

Hence proved.

Obtain Taylor's polynomial expansion of the function $f(x) = \sin x$ about the point $x = \pi/4$. Show that the error term tends to zero as $n \to \infty$ for any real x. Hence, write the Taylor's series expansion of $f(x)$.

(iii)

Given:

$f(x) = \sin x$ about the point $x = \pi/4$

Taylor's series expansion:

$$f(x+a) = f(a) + x f'(a) + \frac{x^2}{2!} f''(a) + \frac{x^3}{3!} f'''(a) + \frac{x^4}{4!} f^{iv}(a) + \cdots$$

Rough

$x = \pi/4$

$x - \pi/4 = 0$

$f(x+a) = f(x)$

$f(x - \pi/4 + \pi/4) = f(x)$

$x = x - \pi/4$

$a = \pi/4$

Taylor expansion of $f(x) = \sin x$ about point $x = \pi/4$

$$f(x) = f(\pi/4) + (x - \pi/4) f'(\pi/4) + \frac{(x-\pi/4)^2}{2!} f''(\pi/4) + \frac{(x-\pi/4)^3}{3!} f'''(\pi/4) + \frac{(x-\pi/4)^4}{4!} f^{iv}(\pi/4) + \cdots$$

$f(x) = \sin x$

$f(\pi/4) = \sin \pi/4 = \frac{1}{\sqrt{2}}$

$f'(x) = \cos x$

$f'(\pi/4) = \cos \pi/4 = \frac{1}{\sqrt{2}}$

$f''(x) = -\sin x$

$f''(\pi/4) = -\sin \pi/4 = -\frac{1}{\sqrt{2}}$

$f'''(x) = -\cos x$

$f'''(\pi/4) = -\cos \pi/4 = -\frac{1}{\sqrt{2}}$

$f^{iv}(x) = \sin x$

$f^{iv}(\pi/4) = \sin \pi/4 = \frac{1}{\sqrt{2}}$

$$f(x) = f(\pi/4) + (x - \pi/4) f'(\pi/4) + \frac{(x-\pi/4)^2}{2!} f''(\pi/4) + \frac{(x-\pi/4)^3}{3!} f'''(\pi/4) + \frac{(x-\pi/4)^4}{4!} f^{iv}(x) + \cdots$$

$$f(x) = \frac{1}{\sqrt{2}} + (x - \pi/4)\left(\frac{1}{\sqrt{2}}\right) + \frac{(x-\pi/4)^2}{2}\left(-\frac{1}{\sqrt{2}}\right) + \frac{(x-\pi/4)^3}{6}(-1/\sqrt{2}) + \frac{(x-\pi/4)^4}{24}(1/\sqrt{2}) + \cdots$$

$$\boxed{f(x) = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}(x - \pi/4) - \frac{1}{2\sqrt{2}}(x - \pi/4) - \frac{1}{6\sqrt{2}}(x - \pi/4)^3 + \frac{1}{24\sqrt{2}}(x - \pi/4)^4 + \cdots}$$

Hence, this is a Taylor expansion of $f(x) = \sin x$ at point $x = \pi/4$.

Now,

Error Terms

$$R_n = \frac{x^{n+1}}{(n+1)!} f^{n+1}(x + \theta a)$$

Given ; $n \to \infty$

$a = \pi/4$

$x = (x - \pi/4)$

$$R_n = \frac{(x - \pi/4)^{n+1}}{(n+1)!} f^{n+1}(x - \pi/4 + \theta \pi/4)$$

$$= \frac{(x - \pi/4)^{\infty}}{(\infty)!} f^{n}\left(x - \pi/4 + \frac{\theta \pi}{4}\right)$$

$$\boxed{R_{n \to \infty} = 0}$$

$\left[\frac{1}{\infty!} = 0\right]$

Hence, Error terms tends to zero as $n \to \infty$.

Sol: Discuss the applicability of Roll's theorem to the function

$$f(x) = |x| \quad \text{in } [-1, 1].$$

→ To verify Roll's theorem

① function should
- Continuous on $[a, b]$
- differentiable on (a, b)

② $f(a) = f(b)$

③ $f'(c) = 0$ such that $c \in (a, b)$.

Condition; ①

Every differentiable function is continuous

Method - I

$$f(x) = |x| \quad \text{in } [-1, 1]$$

$$f(x) = \begin{cases} x & ; \quad x \geq 0 \\ -x & ; \quad x < 0 \end{cases}$$

To verify differentiability;

$$\lim_{h \to 0} \frac{f(a-h) - f(a)}{-h} = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} = f(a)$$

LHD; $\lim_{h \to 0} \frac{f(a-h) - f(a)}{-h}$

$$= \lim_{h \to 0} \frac{f(0-h) - f(0)}{-h}$$

$$\underset{h \to 0}{Lt} \frac{f(-h) - f(0)}{-h}$$

$$\underset{h \to 0}{Lt} \frac{-(-h) - 0}{-h} = \frac{h}{-h} = -1$$

RHD; $\underset{h \to 0}{Lt} \frac{f(a+h) - f(a)}{h}$

$$\underset{h \to 0}{Lt} \frac{f(0+h) - f(0)}{h}$$

$$\underset{h \to 0}{Lt} \frac{h - 0}{h} = 1$$

LHD $\neq$ RHD

$\therefore$ function is not differentiable

$\therefore$ the function $f(x) = |x|$ is not applicable for Rolle's theorem

~~Method-2.~~ ~~To check differentiability~~

~~$f(x) = |x$~~

Verify Rolle's theorem for $f(x) = e^x \sin x$ in $(0, \pi)$.

To verify Rolle's theorem;

1. function should:
   - Continous on $[a,b]$
   - Differentiable on (a,b)
2. $f(a) = f(b)$
3. $f'(c) = 0$ such that $c \in (a,b)$.

Condition

1. we know exponential and trigonometric function is continous and differential

$\therefore$ $e^x \sin x$ also continous and differentiable

2. $f(a) = f(b)$

$f(0) = e^0 \sin 0$

$f(0) = 0$

$f(\pi) = e^\pi \sin \pi$ $\quad (\sin \pi = 0)$

$f(\pi) = 0$

$f(0) = f(\pi)$

3. $f'(c) = 0$

$f(c) = e^c \sin c$

$f'(c) = e^c \cos c + e^c \sin c$

$f'(c) = 0$

$e^c (\cos c + \sin c) = 0$

$\sin c + \cos c = 0$

$\sin c = -\cos c$

$\frac{\sin c}{\cos c} = -1$

$\tan c = -1$ $\quad$ $[\tan c = \tan 3\pi/4]$

$[\tan(3\pi/4) = -1]$

$\boxed{\text{at } c = \frac{3\pi}{4}}$

$c = \frac{3\pi}{4} \in (0, \pi)$

Hence its verify Rolle's theorem at $x = \frac{3\pi}{4}$

$\frac{1}{4} + \frac{1}{4} - 2 - 2$

Verify Rolle's theorem for the function

$f(x) = 2x^3 + x^2 - 4x - 2.$

a polynomial function is continous and differentiable at every point.

But to identify the interval

$f(x) = 0$

$2x^3 + x^2 - 4x - 2 = 0$

at $x = \frac{-1}{2}$ the equation become 0

$\therefore$ $x + \frac{1}{2}$ is the factor of $f(x)$

$$\begin{array}{r|l} x+\frac{1}{2} & 2x^3 + x^2 - 4x - 2 \quad (2x^2 - 4 \\ & \underline{2x^3 + x^2} \\ & -4x - 2 \\ & \underline{-4x - 2} \\ & 0 \end{array}$$

$\left(x + \frac{1}{2}\right)(2x^2 - 4) = 2x^3 + x^2 - 4x - 2$

$(x + 1/2)(2x^2 - 4) = 0$

$x = -1/2$ or $x = \sqrt{2}$ or $x = -\sqrt{2}$

$\therefore$ interval $= [-\sqrt{2}, \sqrt{2}]$

To verify;

- $f(a) = f(b)$

$f(-\sqrt{2}) = (-\sqrt{2})^3 + (-\sqrt{2})^2 + 4(\sqrt{2}) - 2$

$= -4\sqrt{2} + 2 + 4\sqrt{2} - 2$

$f(-\sqrt{2}) = 0$

$f(\sqrt{2}) = (\sqrt{2})^3 + (\sqrt{2})^2 - 4(\sqrt{2}) - 2$

$= 4\sqrt{2} + 2 - 4\sqrt{2} - 2$

$f(\sqrt{2}) = 0$

$f(-\sqrt{2}) = f(\sqrt{2})$

- $f'(c) = 0$ such that $c \in (a, b)$

$f(c) = 2c^3 + c^2 - 4c - 2$

$f'(c) = 6c^2 + 2c - 4$

$f'(c) = 0$

$6c^2 + 2c - 4 = 0$

$3c^2 + c - 2 = 0$

$3c^2 + 3c - 2c - 2 = 0$

$3c(c+1) - 2(c+1) = 0$

$(c+1)(3c-2) = 0$

$c = -1$ or $c = 2/3$

Hence both value of c satisfy the interval

$$\boxed{\begin{aligned} c &= -1 \in (-\sqrt{2}, \sqrt{2}) \\ c &= \frac{2}{3} \in (-\sqrt{2}, \sqrt{2}) \end{aligned}}$$

Hence, its verify our's theorem.